

Answer **all** questions.

1. The information plate from a hairdryer is shown below.

230 V	50 Hz ac
2300 W	

- (a) (i) Use the information in the box to complete the sentences below.

I. Mains electricity has a voltage of [1]

II. The power of the hairdryer is [1]

- (ii) Use your answers to (i) and the equation:

$$\text{current} = \frac{\text{power}}{\text{voltage}}$$

to calculate the current in the hairdryer. [2]

current = A

- (b) (i) A vacuum cleaner has a current of 2 A.

Circle the correct value for the fuse that should be used in the vacuum cleaner. [1]

1 A 3 A 13 A

- (ii) Tick (✓) the box that correctly states the purpose of a fuse: [1]

Fuses are safety features designed to prevent electric shocks

☐

Fuses are safety features designed to prevent fires

☐

Fuses are safety features designed to prevent explosions

☐


Examiner
only

(iii) Circle the device that has the same purpose as a fuse.

[1]

earth wire

miniature circuit breaker

residual current circuit breaker

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